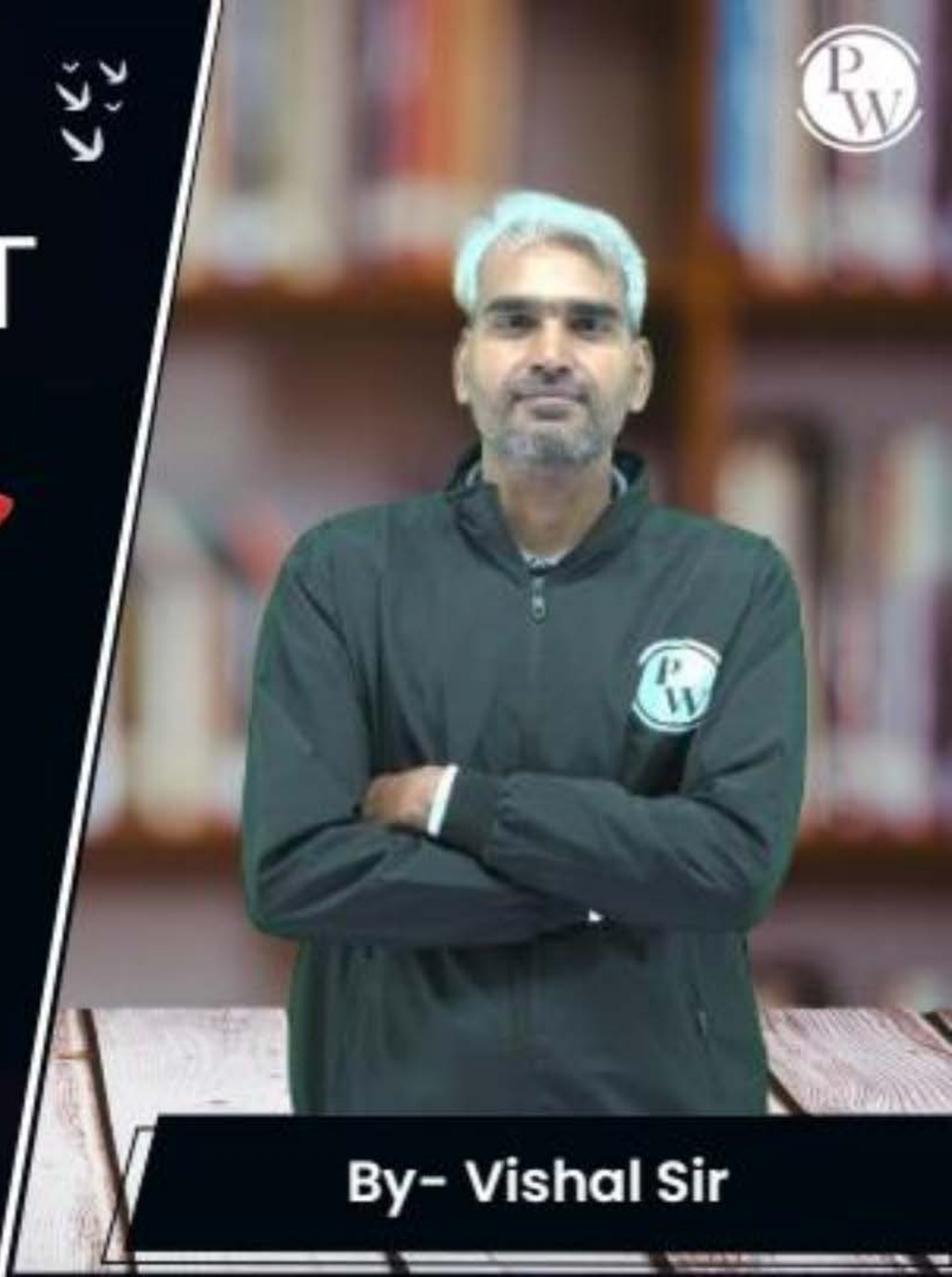
Computer Science & IT



Database Management System

Query Languages

Lecture No. 06



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Recap of Previous Lecture



- **Topic** SQL commands
- **Topic** SQL clauses
- **Topic** Order of execution

Slide

Topics to be Covered



- **Topic** Nested queries (Sub queries)
- **Topic** IN, ANY, ALL and EXISTS operators
- **Topic** Execution of Nested queries

Slide



Topic : Nested queries

Nested Query (Query within query)

Independent Nested query

When inner query does not depend upon Outer query
{i.e. it can be executed independently}

Correlated Nested query

Inner query uses the attribute from the relation specified in the Outer query



Topic : Independent nested query

eg: Select Sid
From Student
Where (marks = (Select max(marks)
From Student))

Outer query

inner query

it can be executed independently
i.e. Independent query

Note: If inner query is independent query then, inner query will be executed first.



Topic : Correlated nested query

Select *
From R
Where some operator (Select *
From S
Where (R.A = S.B))

outer query

inner query

because of this it is a Correlated query



Topic : Correlated nested query

Order of Execution

6 — Select *

1 — From R

5 — Where some operator (Select * — 4

From S — 2

Where (R.A = S.B) — 3

for each tuple of Outer relation(R) this three steps will be repeated w.r.t. Complete inner relation(S)

NOTE :- ① Best used operators w.r.t. independent nested query are IN, ANY, ALL

② Best used operator w.r.t. Correlated nested query is "EXISTS"



Topic : IN operator

- IN operator is used to check whether the tuple under consideration is present in a given set of tuples or not.

tuple under consideration

Set of tuples

eg: X IN {2, 4, 6, 8, 9}

if X = 4, then IN will return true

if X = 5, then IN will return false

Complement of IN is "NOT IN"

Q:

Supplier (Sid, Sname, Rating)
Parts (Pid, Pname, Color)
Catalog (Sid, Pid, Cost)

Catalog

S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_1

Parts

P_1	Red
P_2	Green
P_3	Red

Retrieve Sids of the suppliers who supplied some Red Color Parts

S_1, S_3

Soluⁿ:

```
Select Sid
From Catalog
Where Pid IN { Select Pid
               From Parts
               Where (Color = 'RED') }
```

{ P_1, P_3 }

Output of inner query will provide us the set of tuples

* If inner query result is empty (i.e. set of tuples is empty) then IN operator will always return false

"NOT IN" is Complement of IN

∴ "NOT IN" will return true if inner query result is EMPTY



Topic : ANY operator

ANY operator is used along with other Comparision Operators
{i.e. =, <, >, ≤, ≥, <> }
↑ Not equal

"ANY" operator will return true if at least one tuple in set of tuples satisfy the Comparison Condition.

Eg: $X \leq ANY \{2, 4, 6, 8, 9\}$
(Comparison Condition ← set of tuples)

if X=5, then ANY operator will return true

if X=15, then ANY operator will return false

Note: "=ANY" is equivalent to "IN" operator

Note:- If inner query result is empty then there is no tuple in the set of tuples to satisfy the Comparision Condition

∴ If inner query result is empty then ANY operator will always return false



Topic : ALL operator

- Just like "ANY" operator, "ALL" operator is also used along with other Comparision Condition
- ALL operator returns false only if at least one tuple in the set of tuples fails the comparison condition
  - ∴ If there is no tuple in the set of tuples then "ALL" operator will always return true

eg: X >ALL {2, 4, 6, 8, 9} set of tuples

if X = 20, then ALL returns true

if X = 7, then ALL will return false

Note: "NOT IN" is equivalent to "<>ALL"



Topic : EXISTS operator

- If inner query result is non-empty then "EXISTS" operator will return true

- If inner query result is Empty then "EXISTS" operator will return false

- Complement of "EXISTS" is "NOT EXISTS"

H.W.

#e.g.

```
SELECT C.sid
FROM Catalog C
WHERE EXISTS ( SELECT *
               FROM Parts P
               WHERE (P.Pid=C.Pid AND P.color='RED'))
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_3

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

What will be the output of above query if it is executed W.r.t given instances of Catalog & Parts.

H.W.

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts

B. Sids of suppliers who supplied only proper subset of parts from all parts

C. Sids of suppliers who supplied all parts

D. Sids of suppliers who did not supply any part



2 mins Summary

Topic Nested queries

Topic IN, ANY, ALL and EXISTS operators

Topic Execution of Nested queries

Slide

THANK - YOU